Occupancy-matched PC1 composite

(a) Near-zero PC1
238 months | Amplitude = 0.147 | η = 0.147

(b) Most positive PC1
199 months | Amplitude = 0.505 | η = 0.094

(c) Most negative PC1
79 months | Amplitude = 0.844 | η = 0.092

CAE cluster composite

(d) Cluster 1 (Low-amplitude SAM−)
238 months | Amplitude = 0.162 | η = 0.280

(e) Cluster 2 (SAM+)
199 months | Amplitude = 0.473 | η = 0.064

(f) Cluster 3 (SAM−)
79 months | Amplitude = 0.770 | η = 0.060



